
Assumptions: EV is charged at home

Gasoline price = \$3.62/gallon

Electricity price = \$0.18/kWh

CO₂ grid intensity = ~~91.424 lb CO₂/mWh~~
= ~~0.4184 kg CO₂/kWh~~

ICE fuel economy = ~~27.1 mpg~~

Hybrid fuel economy = 50 mpg

EV energy use = 33 kWh/100mi
= 0.33 kWh/mi

Tailpipe emissions = 9 kg CO₂/gal

For ICE power train

gallons used for the 10,000 miles

1 gal $\rightarrow$ 27.1 mi

x gal $\rightarrow$ 10,000 mi

x gal = $\frac{10,000 \text{ mi} \times x \text{ gal}}{27.1}$

= 369 gal

Use of fuel

\$3.62 $\times$ 369 gal

= \$1,335.78

Emissions

9 kg CO₂ per gal

$$\begin{array}{ccc} 9 \text{ kg} & \times & 1 \text{ gal} \\ x \text{ kg} & \Rightarrow & 269 \text{ gal} \end{array}$$

$$= 3,321 \text{ kg CO}_2$$

Hybrid power train

1 gal $\rightarrow$ 50 mi

x gal $\rightarrow$ 10,000 mi

$$x \text{ gal} = \frac{10,000 \text{ mi}}{50 \text{ mi/gal}} = 200 \text{ gal}$$

Fuel cost

$$\begin{array}{ccc} \$ 8.62 & \times & 1 \text{ gal} \\ \$ x & \Rightarrow & 200 \text{ gal} \end{array}$$

$$\$ x = \$ 1724$$

CO₂

$$\begin{array}{ccc} 9 \text{ kg CO}_2 \text{ eq} & \Rightarrow & 1 \text{ gal} \\ x & \Rightarrow & 200 \text{ gal} \end{array}$$

$$= \underline{\underline{1,800 \text{ kg CO}_2}}$$

Electric Vehicle Power from
Electricity Consumption

$$\begin{aligned} 0.39 \text{ kWh} &\rightarrow 1 \text{ mi} \\ \times &\quad \rightarrow 10,000 \\ &= 3,900 \text{ kWh} \end{aligned}$$

Cost of energy (Charging at home)

$$\begin{aligned} \$0.18 &\rightarrow 1 \text{ kWh} \\ \times &\rightarrow 3,900 \text{ kWh} \\ &= \$702 \end{aligned}$$

Emissions

$$\begin{aligned} 0.4134 \text{ kg CO}_2 &\rightarrow 1 \text{ kWh} \\ \times &\quad \rightarrow 3,900 \text{ kWh} \\ &= 1,612.3 \text{ kg CO}_2 \end{aligned}$$

Savings

cost:	ICE	\$ 1,335.78
	hybrid	\$ 724.00
		\$ 611.78

$$= 611.78 \leq \$612$$

$$\begin{array}{r} \text{ICE} - \$1325.78 \\ \text{EV} - 702.00 \\ \hline \$633.78 \\ \hline \approx \$634 \end{array}$$

CO₂ Emission savings

$$\begin{array}{r} \text{ICE} \quad 3321 \\ \text{hybrid} \quad 1800 \\ \hline 1,521 \text{ kg CO}_2 \end{array}$$

$$\begin{array}{r} \text{ICE} \rightarrow 3321.0 \\ \text{EV} \rightarrow 1612.3 \\ \hline 1,708.7 \text{ kg CO}_2 \end{array}$$

This information shows that the EV option offers more economical and environmental benefit. Since it doesn't have tail pipe emissions however, it is important to note that if not being home charged, cost of energy might change from vendor to vendor. Also, the environmental impact of batteries used in both EV & hybrid was not considered.

Since the lifecycle phase considered here is only the use phase.

If charging is not done at home, the hybrid option offers a good competitive advantage over the EV. While the savings may vary slightly for EV and hybrid over the given mileage, longer use may show significant savings over time all things being equal.

